

Survey for GraphRAG-Driven Modeling of Large-Scale Multimodal Data with X-Agent Simulation for Insights

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Abstract The increasing complexity of **social media** ecosystems presents a critical **Big Data** challenge: how to efficiently filter signal from noise and generate actionable, multi-perspective insights without incurring prohibitive computational overhead. This paper surveys and synthesizes a constellation of recent advances in multi-agent simulation, knowledge graph engineering (KG), and retrieval-augmented generation that address the *three V's* of Big Data—volume, velocity, and variety.

From a **data volume and variety** perspective, the fusion of multimodal signals (text, metadata, interaction graphs) is essential for coherent situational awareness. Foundational work on multimodal data fusion provides the theoretical underpinnings for integrating heterogeneous streams, while temporal knowledge graph frameworks such as Graphiti enable the construction of evolving to capture the dynamics of public discourse without full-dataset recomputation. Complementing this, graph embedding techniques like Node2Vec combined with FAISS indexing allow sub-linear similarity search over high-dimensional entity representations.

Addressing the **velocity** dimension, agentic frameworks such as **EvoCorps** and **BettaFish** model online interactions as multi-agent Markov decision processes. By simulating role-specialized agents in closed-loop evolutionary dynamics, these systems move beyond static statistical inference to anticipatory, population-level simulation. The **MARS** framework further optimizes this process by introducing budget-aware multi-agent reinforcement learning, balancing insight depth against computational expenditure during deep research tasks.

To ensure **insight quality** amidst noisy, high-volume data streams, recent advances in Graph Retrieval-Augmented Generation (GraphRAG) provide structured reasoning capabilities. Unlike purely semantic retrieval, **GraphRAG** leverages entity-relation topologies for multi-hop inference, while iterative retrieval strategies interleaved with chain-of-thought reasoning and **GNN** scoring improve the recall and ranking of bridge evidence crucial for complex query answering. Domain-adaptive fine-tuning pipelines, exemplified by **EasyDataset** and **LLaMA-Factory**, further ground language models in specialized regulatory and social contexts, reducing hallucination and improving factual reliability.

Collectively, these technologies—spanning temporal knowledge representation, multi-agent social simulation, and attention-guided graph retrieval—constitute a coherent toolkit for transforming deluge of social media data into structured, explainable, and low-latency intelligence. This synthesis underscores a paradigm shift from post-hoc analytics to *in-process, closed-loop insight generation*, where user attention dynamics are explicitly modeled as a high-value filter to prune the analytical

Keywords

Big Data Analytics, User Attention Modeling, Graph Retrieval-Augmented Generation (Graph-RAG), Knowledge Graph (KG), Digital Twin Simulation, Social Media Mining, Semantic Entropy Quantization, Temporal Graph Reasoning

Introduction

1.1 Big Data Challenge in Digital Public

The digital public sphere has evolved into a high-velocity, high-volume information ecosystem where public relations (PR) crises emerge. The ability to forecast narrative shifts is not a simple keyword monitoring. It constitutes a formidable **Big Data challenge**, classically characterized by the *three V's* [1, 2]:

1. **Volume:** Social platforms generate millions of user comments, shares, and sentiment signals daily.
2. **Velocity:** Public opinion shifts driven by amplification or algorithmic feedback loops. Processing data hours or days after an event—are rendered ineffective for proactive intervention [3].
3. **Variety:** Fusing these disparate signals to form a coherent situational picture remains a non-trivial engineering problem [4, 5].

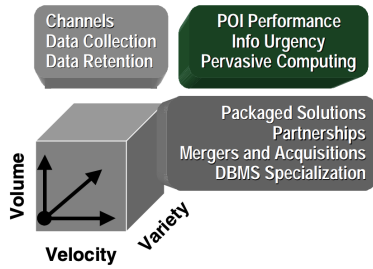


Figure 1. 3V-Dimension Big Data Challenges.

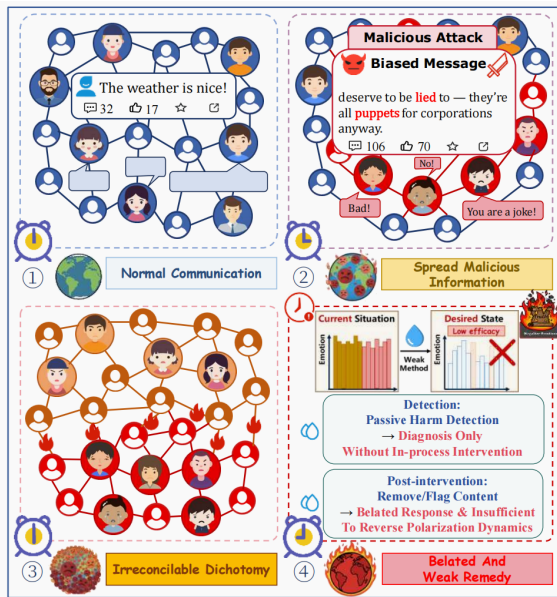


Figure 2. Online Interaction among Virtual Agents. The figure illustrates that communication can evolve from normal user interactions into malicious behavior, where harmful information is spread by adversarial agents. This process amplifies negative emotions and leads to increased social polarization. Existing harm detection and post-hoc interventions.

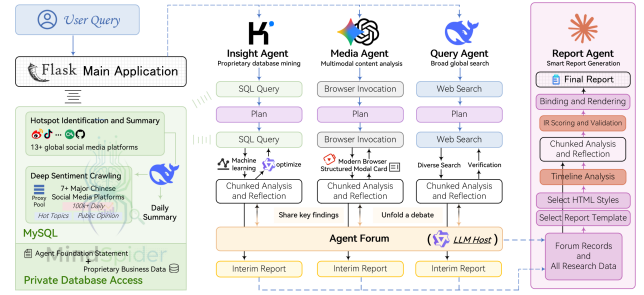


Figure 3. BettaFish Framework: The overall architecture consists of multiple specialized agents, including Insight Agent for private database mining, Media Agent for multimodal content analysis, Query Agent for precise web search, and Report Agent for structured report generation.

To cope with this complexity, recent research has turned to **agentic frameworks**—multi-agent systems powered by large language models (LLMs) to simulate social dynamics. Architectures such as **BettaFish** and **EvoCorps** [6, 7] show potential of role-specialized agents.

1.2 Limitations of Existing Agentic Frameworks

First, they suffer from *weak user-demand modeling*. Existing frameworks treat all data with equal weight, retrieving over vast document corpora without a principled mechanism to filter content based on what the audience *actually cares about*. This results in low-relevance, high-entropy content.

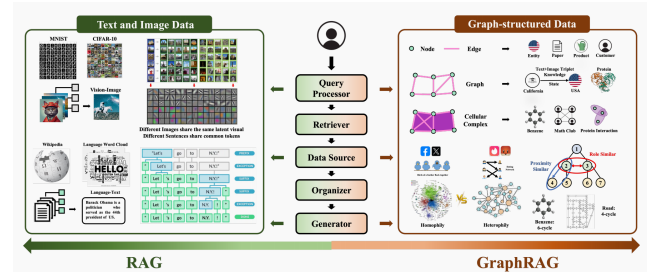


Figure 4. Comparison between RAG and GraphRAG: RAG typically operates on text or image data, which can be flattened into 1D sequences or 2D grids, but lacks explicit relational structure. As a result, it struggles to model complex dependencies and domain-specific relationships in data.

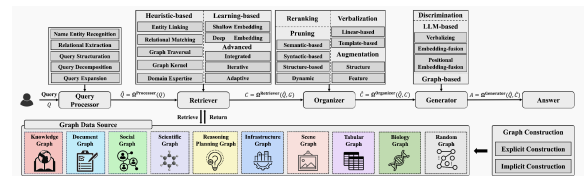


Figure 5. GraphRAG Framework Components.

Second, they rely on *static reasoning graphs*. Once a knowledge graph is constructed or a retrieval pool is populated, the system's attention is fixed. It fails to adapt dynamically to the temporal evolution of public sentiment or the shifting focus of user attention. Consequently, the state space of these simulations grows exponentially with the number of agents and interaction depth. In **MiroFish**-like architectures, the state complexity is characterized

by a large search width and depth (standard configuration $rolls = 60$, $k = 56$ min interaction steps), leading to prohibitive token consumption and latency. These need a paradigm shift toward **attention-anchored computation**—a methodology that leverages real-time signals of user focus to prune the analytical search space.

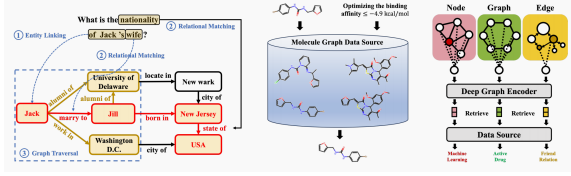


Figure 6. Representation Retriever in GraphRAG

From a Big Data engineering standpoint is two-fold: **insight quality** and **computational economy**.

Methods

The methodology aimed to addresses limitations of prior agentic, weak user-demand modeling and static reasoning chains—by explicitly incorporating user-attention priors to compress the analytical search space.

2.1 Architectural Basis: MulAgent Simulation

The frame of EvoCorps, formulates online discourse governance as a dynamic socials between an intervention team and a population of ordinary users [7]. In **EvoCorps (2026)**, environment is modeled as a **Multi-Agent Markov Decision Process (MMDP)** defined by tuple:

$$\mathcal{M} = (\mathcal{N}, \mathcal{S}, \{\mathcal{A}^i\}_{i \in \mathcal{N}}, P, R) \quad (1)$$

where $\mathcal{N} = \{\text{Analyst, Strategist, Leader, Amplifier}\}$ denotes the set of heterogeneous agents, $\mathcal{S}$ is the state space, $\mathcal{A}^i$ is the action space of agent i , P is the state transition function, and R is the reward function. The state $s_t = (v_t, e_t)$ is a mean-field summary of the discourse environment, aggregating opinion extremity v_t and sentiment e_t over the user population [7].

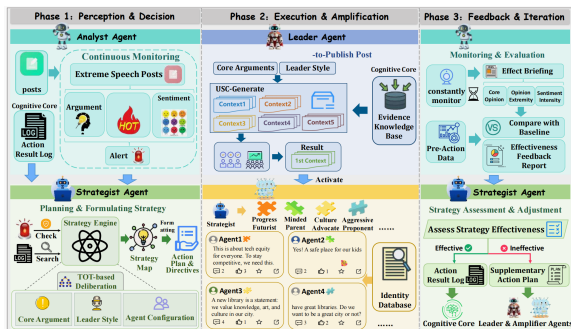


Figure 7. Closed-loop Evolution Flow in EvoCorps: The framework consists of perception and decision, execution and amplification, and feedback and iteration stages, enabling adaptive strategy optimization through multi-agent coordination.

The intervention team executes a joint action $a_t = (a_t^{\text{stg}}, a_t^{\text{ldr}}, a_t^{\text{amp}})$ conditioned on a retrieval-augmented collective cognition core, which maintains an evidence knowledge base K_t and an action-outcome memory M_t :

$$a_t \sim \pi(a_t | s_t, K_t, M_t) \quad (2)$$

Learning objective maximizes cumulative reward $J = \mathbb{E}_\pi \left[\sum_{t=1}^T R(s_t, a_t) \right]$ through evolution updates to K_t and M_t rather than gradient-based optimization of model parameters [7].

2.2 Domain-Adaptive Model Fine-Tuning

Compared to **General-purpose LLMs**, trained on broad web corpora, lack the nuanced understanding of platform-specific regulations, legal frameworks, and historical case law required for high-fidelity PR simulation. To mitigate this, we employs **supervised fine-tuning** on a curated domain corpus $\mathcal{D}_{\text{domain}}$ comprising high-quality QA pairs [8], regulatory documents, and case analyses.

Given a base model parameterized by θ_0 , the fine-tuning objective has standard causal language modeling formula:

$$\mathcal{L}_{\text{ft}}(\theta) = -\mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{domain}}} \left[\sum_{t=1}^{|y|} \log P(y_t | x, y_{<t}; \theta) \right] \quad (3)$$

where x denotes the instruction or context, and y the target response. The dataset $\mathcal{D}_{\text{domain}}$ is constructed using the EasyDataset [8] augmentation toolkit, which generates synthetic but contextually grounded QA pairs from seed documents [8]. Fine-tuning is executed via the LLaMA-Factory pipeline with LoRA [9].

The output of this stage is a domain-adapted model θ_{domain} that internalizes the legal and social priors of the target environment.

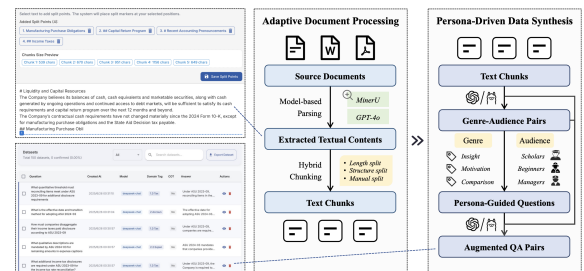


Figure 8. EasyDataset Pre-Processing Framework: Data processing tool for efficiently building, cleaning, and enhancing high-quality training data (such as QA pairs) to support large model fine-tuning and data-driven tasks.

	LLaMAFactory	FastChat	LitGPT	LMFlow	Open-Instruct
LoRA	✓	✓	✓	✓	✓
QLoRA	✓	✓	✓	✓	✓
DoRA	✓				
LoRA+	✓				
PiSSA	✓				
GaLore	✓	✓		✓	✓
BAdam	✓				
Flash attention	✓	✓	✓	✓	✓
S ² attention	✓				
Unsloth	✓		✓		
DeepSpeed	✓	✓	✓	✓	✓
SFT	✓	✓	✓	✓	✓
RLHF	✓			✓	
DPO	✓				✓
KTO	✓				
ORPO	✓				

Table 1. Comparison Features of LLaMAFactory with Popular Frameworks for Fine-Tuning LLMs: compares LLaMA-Factory with mainstream LLMs fine-tuning frameworks FastChat, LitGPT, LMFlow, and Open-Instruct, highlighting its more comprehensive feature support across multiple dimensions, including efficient fine-tuning algorithms like DoRA and PiSSA, performance optimization operators like Unsloth and S² attention, and alignment training protocols such as KTO and ORPO.

	Freeze-tuning	GaLore	LoRA	DoRA	LoRA+	PiSSA
Mixed precision	✓	✓	✓	✓	✓	✓
Checkpointing	✓	✓	✓	✓	✓	✓
Flash attention	✓	✓	✓	✓	✓	✓
S ² attention	✓	✓	✓	✓	✓	✓
Quantization	×	×	✓	✓	✓	✓
Unsloth	×	×	✓	✓	✓	✓

Table 2. Compatibility between the fine-tuning techniques featured in LLaMAFactory

2.3: User-Focus Prior and MulAgent Retrieval

The explicit modeling of *user attention* as a prior to guide and constrain the retrieval process. This stage is implemented as the **MARDS (MARS + Deep Search)** architecture, which extends the **CAIR-MARS** framework [19] with deep, iterative search capabilities.

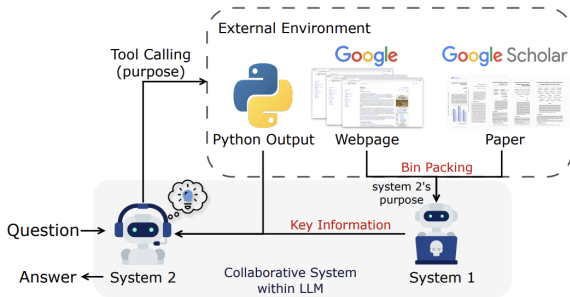


Figure 9. MARD Dual-System Collaborative Frame: it demonstrates a co-evolutionary system based on LLMs, with System 1 and System 2: System 2 receives the problem and invokes external tools like Python, web search, and System 1 processes and compresses the massive amount of raw data using Bin Packing technology, ultimately feeding back key information to System 2 to generate the final answer.

2.3.1 Attention-Aware Query Structuring

Let Q denote user’s query. A query processor Ω^{Proc} performs named entity recognition and relation extraction

to decompose Q into a set of structured sub-queries $\{q_1, \dots, q_m\}$, each linked to entities in a domain-specific knowledge graph. Concurrently, the Google Hypothesis module[11] crawls web sources in real time to construct an *attention pool*:

$$\mathcal{A}_Q = \{(u_i, s_i)\}_{i=1}^K \quad (4)$$

where u_i is a URL or content snippet and s_i is an initial relevance score via BM25 or embedding similarity.

2.3.2 Semantic Entropy-Controlled GraphRAG

A challenge in RAG is the retrieval of low-quality or hallucinatory content, particularly when the retriever relies solely on embedding similarity [15]. To address this, it introduces **semantic entropy quantization** as a filtering mechanism. For a candidate document chunk d retrieved in response to sub-query q , we compute its semantic entropy with respect to the attention pool $\mathcal{A}_Q$:

$$H(d | \mathcal{A}_Q) = - \sum_{a \in \mathcal{A}_Q} P_{\text{align}}(d | a) \log P_{\text{align}}(d | a) \quad (5)$$

where $P_{\text{align}}(d | a)$ is the probability that chunk d aligns with the factual or narrative content of anchor a , estimated via an LLM-based verifier [15]. A high entropy value indicates that d contains contradictory or ambiguous information relative to the established attention anchors, and such chunks are discarded if $H(d | \mathcal{A}_Q) > \tau_H$.

The retained chunks are then scored using a weighted combination of semantic relevance and attention fidelity:

$$S(d) = \alpha \cdot \phi(\mathbf{e}_q, \mathbf{e}_d) + (1 - \alpha) \cdot (1 - H(d | \mathcal{A}_Q)) \quad (6)$$

where $\phi(\cdot, \cdot)$ denotes cosine similarity between query and doc embeddings, and $\alpha \in [0, 1]$ is a balance hyperparam. The retrieval process itself is iterative and graph-guided. Starting from seed entities identified in Q , the retriever traverses the document graph using a variant of personalized PageRank [14] or GNN-based scoring [14], pruning paths with cumulative scores below a threshold. The quality of this retrieval is evaluated with metrics:

- **Normalized Discounted Cumulative Gain (NDCG)** measures the ranking quality of retrieved chunks, with higher weights assigned to top-ranked items.
- **Mean Reciprocal Rank (MRR)** quantifies position of the first relevant chunk, which for multi-hop reasoning where missing a bridge document early in the chain can derail entire inference [15].

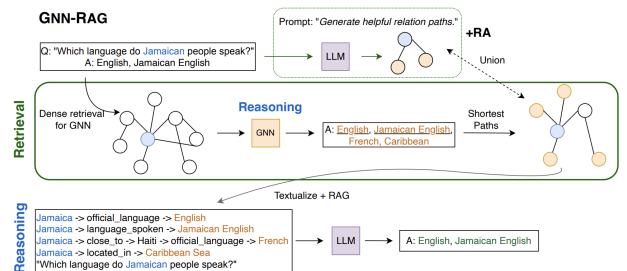


Figure 10. GNN-RAG Framework: GNN reasons for dense subgraph to retrieve answers, while retrieved reasoning paths

During KGQA, the message passing is also conditioned to the given question q . For readability purposes, we present the following GNN update for KGQA:

$$h_v^{(l)} = \psi \left(h_v^{(l-1)}, \sum_{v' \in \mathcal{N}_v} \omega(q, r) \cdot m_{vv'}^{(l)} \right)$$

where $\omega(\cdot)$ measures relevance between relation r of fact (v, r, v') and the query q . Neighbor messages $m_{vv'}^{(l)}$ are aggregated by a sum operator $\sum$ over the neighborhood $\mathcal{N}_v$.

2.4 Digital Twin World and Dynamic Simulate

The third stage constructs a **temporal and causal knowledge graph** that functions as a Digital Twin of the evolving discourse environment. Unlike static knowledge graphs used in conventional RAG system [15], it models the temporal validity, enabling dynamic, non-template simulation of behaviors.

2.4.1 Temporal Knowledge Graph Construction

The graph is constructed using **Graphiti** framework for temporal edge weighting and **Zep** for session-based memory persistence [16]. The graph at time t is a heterogeneous directed graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t, \mathcal{R}, \mathcal{T})$, where:

- $\mathcal{V}_t$ is the set of entity nodes (e.g., streamer, platform, regulation, fan community).
- $\mathcal{E}_t \subseteq \mathcal{V}_t \times \mathcal{R} \times \mathcal{V}_t \times \mathcal{T}$ is the set of typed, timestamped edges.
- $\mathcal{R}$ is the set of relation types (e.g., *violates*, *endorses*, *retracts*, *criticizes*).
- $\mathcal{T}$ assigns a temporal interval $[t_{\text{start}}, t_{\text{end}})$ to each edge, denoting its period of validity.

Each edge is further augmented with a **sentiment weight** $\omega_{\text{sent}} \in [-1, 1]$, derived from fine-grained emotion analysis of the underlying text [16], and a **causal flag** $c \in \{0, 1\}$ indicating whether the relation represents a causal link (e.g., “policy change $\rightarrow$ user backlash”).

2.4.2 Graph Structural Analysis and Embedding

To uncover latent community structure, NEXUS applies the **Louvain algorithm** for community detection. Louvain optimizes the modularity Q of the graph partition:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \quad (7)$$

where A_{ij} is the edge weight between nodes i and j , $k_i = \sum_j A_{ij}$ is the degree of node i , $m = \frac{1}{2} \sum_{i,j} A_{ij}$ is the total edge weight, and $\delta(c_i, c_j)$ is the Kronecker delta function indicating whether nodes i and j belong to the same community c .

Node embeddings are generated using **Node2Vec**, which learns continuous vector representations by maximizing the likelihood of preserving network neighborhoods discovered through biased random walks [13]. Given a source node u , the random walk transition probability is:

$$P(v | u) = \frac{\pi_{uv}}{Z} \quad \text{where} \quad \pi_{uv} = \alpha_{pq}(t, x) \cdot w_{uv} \quad (8)$$

Here, $\alpha_{pq}(t, x)$ is a search bias that interpolates between breadth-first and depth-first sampling, controlled by return parameter p and in-out parameter q . The resulting embeddings $\mathbf{h}_v$ are indexed using **FAISS** with an IVF-PQ.

2.4.3 Attention-Guided Dynamic GraphRAG

To answer complex analytical queries about discourse evolution, we employ an **Attention-guided Dynamic GraphRAG** mechanism. Unlike static subgraph retrieval [?], the retrieval is conditioned on both the query Q and the current Digital Twin state $\mathcal{G}_t$. A Graph Neural Network (GNN) retriever [14] computes node relevance scores via message passing:

$$\mathbf{h}_v^{(l)} = \sigma \left(\mathbf{W}^{(l)} [\mathbf{h}_v^{(l-1)} \parallel \mathbf{q}_{\text{enc}}] + \sum_{u \in \mathcal{N}(v)} \mathbf{W}_{\text{msg}}^{(l)} \mathbf{h}_u^{(l-1)} \right) \quad (9)$$

where $\mathbf{q}_{\text{enc}}$ is the LLM-encoded query representation, $\mathcal{N}(v)$ is the neighbor set of v , and $\parallel$ denotes concatenation. The relevance score for node v after L layers is $r_v = \sigma(\mathbf{w}^\top \mathbf{h}_v^{(L)})$.

The retriever then extracts a reasoning subgraph $\mathcal{G}_{\text{sub}}$ by performing L -hop expansion from seed entities, pruning paths whose cumulative attention score falls below a dynamic threshold τ_{attn} .

2.4.4 Narrative-Based Cognitive Simulation

A distinguishing feature of it is the **Narrative-based Cognitive Driver**. For each stakeholder community identified by **Louvain clustering**, LLM-powered agents with a defined belief system, emotional baseline, and behavioral tendencies [16]. The agent is presented with the pruned reasoning subgraph $\mathcal{G}_{\text{sub}}$ and prompted to generate a *cognitive response*. Simulated cognitive responses:

$$P(A | Q, \mathcal{G}_t) \approx \text{LLM}_{\theta_{\text{dom}}} (A | \text{Verb}(\mathcal{G}_{\text{sub}}), \{\text{Cog}_i\}_{i=1}^C) \quad (10)$$

where $\text{Verbalize}(\cdot)$ converts subgraph into a linearized prompt relational topology, temporal constraints.

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